

**UCLA**

**Samueli**  
School of Engineering

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# GoFish!: A Tension Regulation Fishing Robot

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TEAM 17

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# Overview

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# Project Overview & Motivation

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- **Project Goal**

- Develop a 1-DOF rigid robotic fishing rod
- Simulate fish-fighting interaction under controlled testing
- Develop a control paradigm to regulate fishing-line tension near a 3.5 N target with a  $\pm 0.1$  N deadband.

- **Motivation / Literature**

- Prior fishing haptic devices show that force feedback improves realism in fishing simulation.
- Fishing-rod haptic displays have also been studied for presenting force sensation through a rod-and-line structure.

- **Applications**

- Fishing training simulator
- Entertainment / interactive game controller
- Robotics force-feedback education

- **System Concept**

$$F_{pull} \rightarrow T_f \rightarrow \theta_{cmd} \rightarrow \theta$$

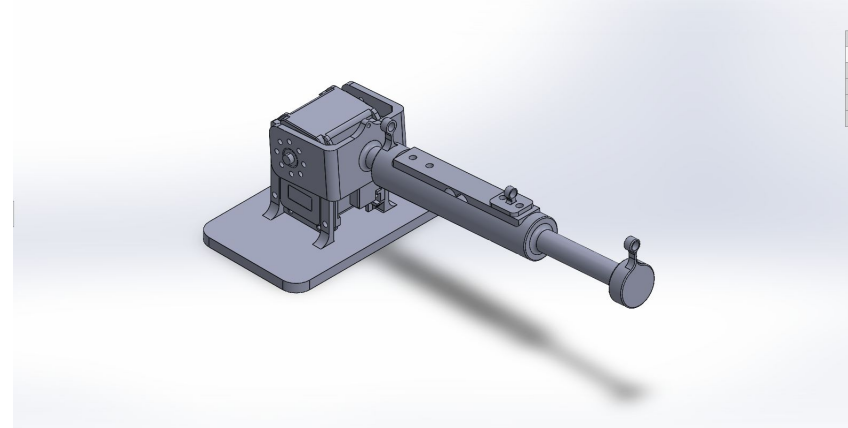
# Hardware

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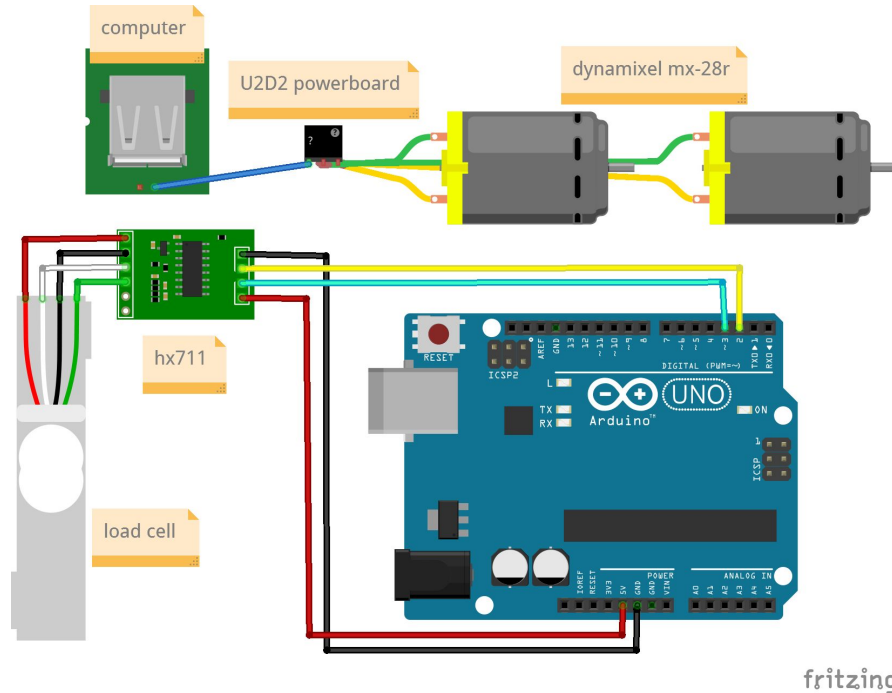
# Hardware Test Video



GoFish: Tension Regulation Fishing Robot



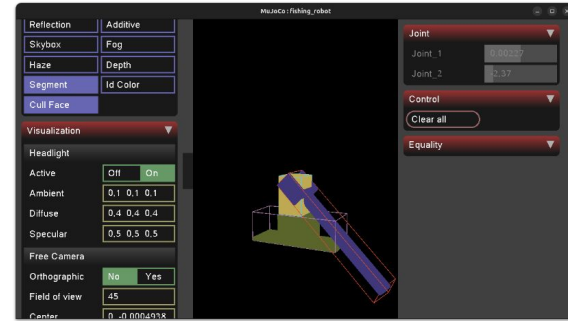
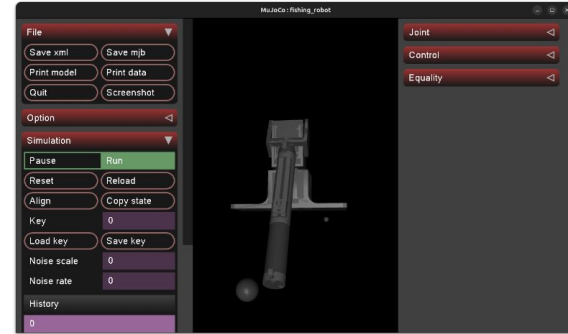
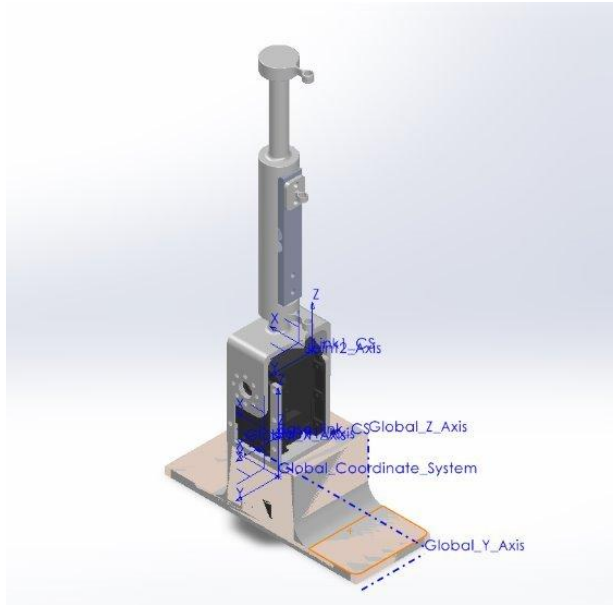
# Wiring Diagram



# Software

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# Simulation (mujoco)





# Codebase (true implementation)

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## ROS2 - software architecture modularity

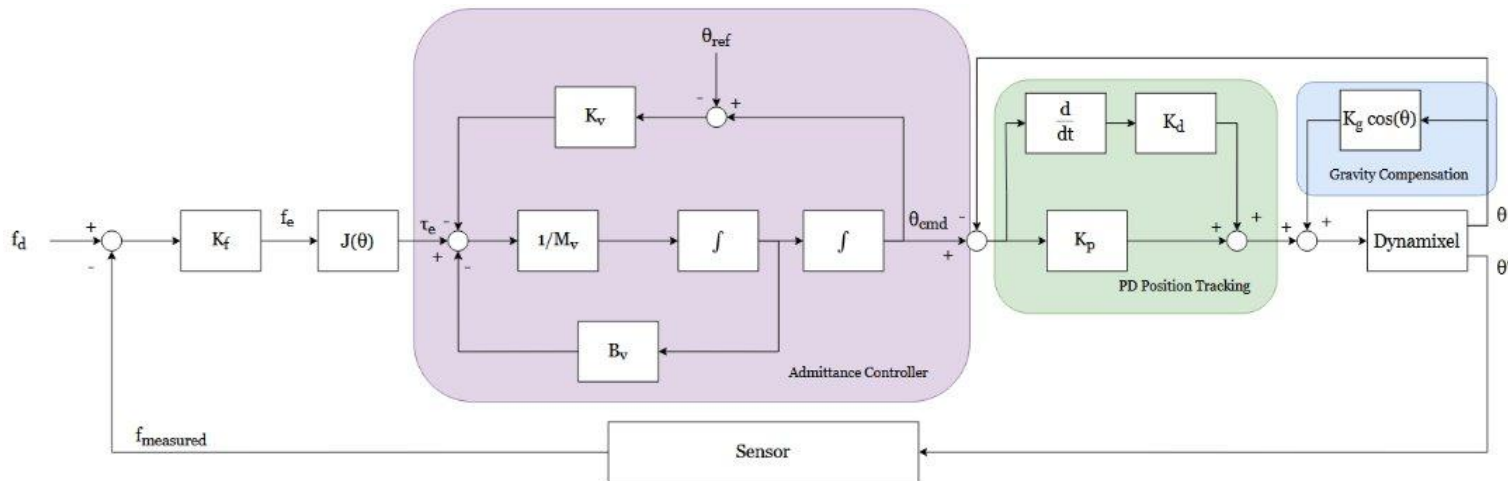
Building out the ROS2 codebase such that the control logic interacts seamlessly with Mujoco simulation and hardware control

- *Bringup*: launch configuration. Can launch simulation or hardware control based on preference
- *Description*: mesh and model descriptors
- *Control*: tension control package
- *Planning*: will likely be used to control randomness of 'fish' behavior. Can be thought of as a fish agent (our disturbance)
- *Sensors*: sensor readings and publishing
- *Interfaces*: develop protocol for inter-node communication

# Controller & Simulation

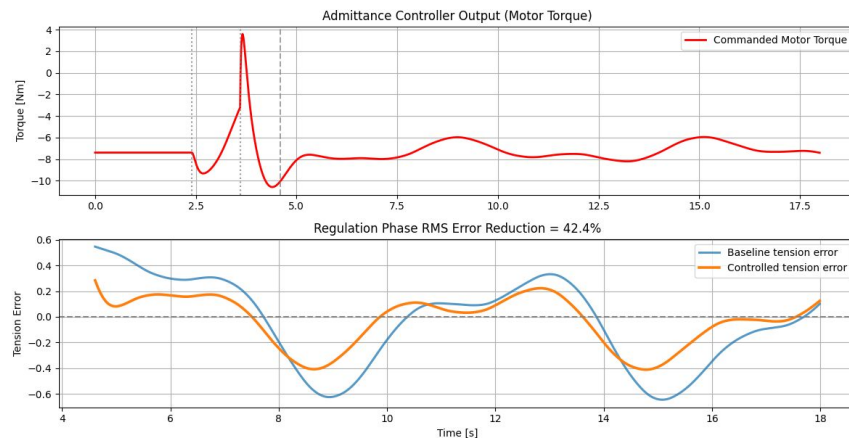
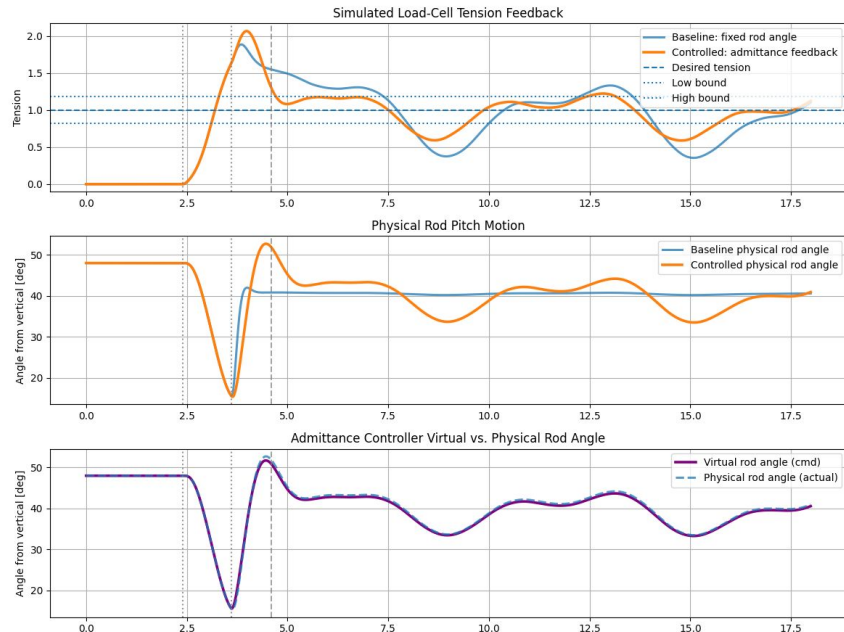
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# Controller Diagram (Admittance Control)

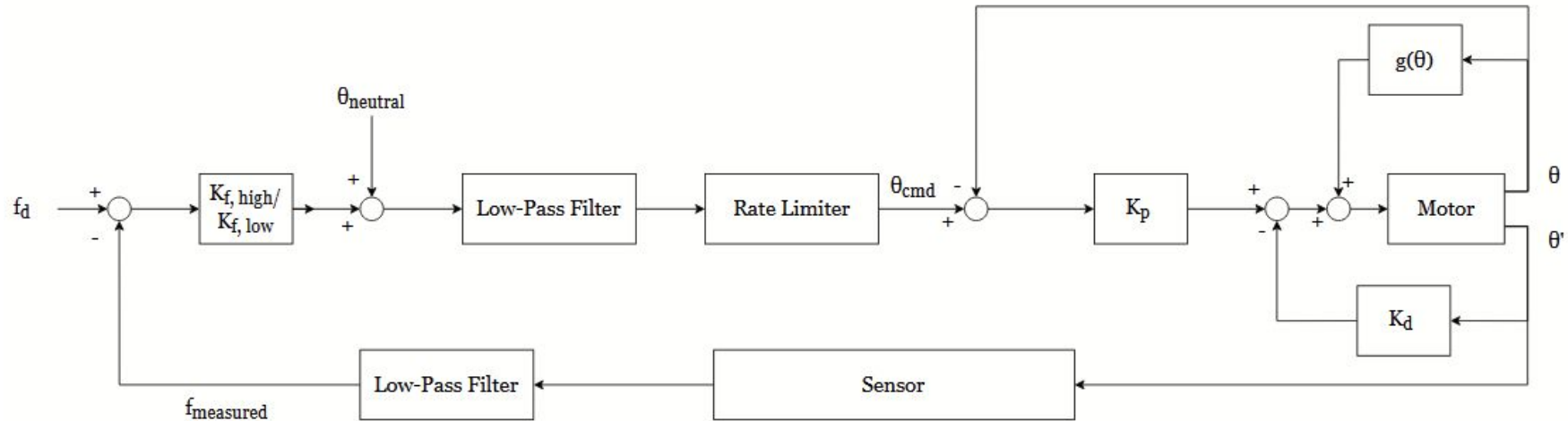


$$\mathbf{M}_v \ddot{\theta}_{cmd} + \mathbf{B}_v \dot{\theta}_{cmd} + \mathbf{K}_v (\theta_{cmd} - \theta_{ref}) = -\mathbf{J}(\theta) \mathbf{K}_f (f_d - f_{measured})$$

# Admittance Controller Results

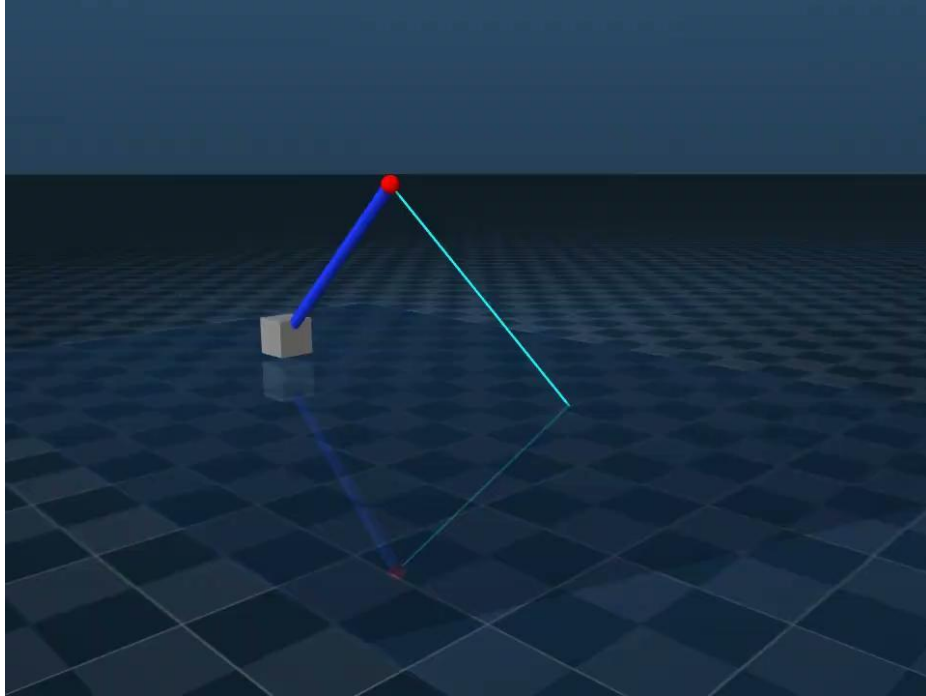


# Controller Diagram (Proportional Force Control)

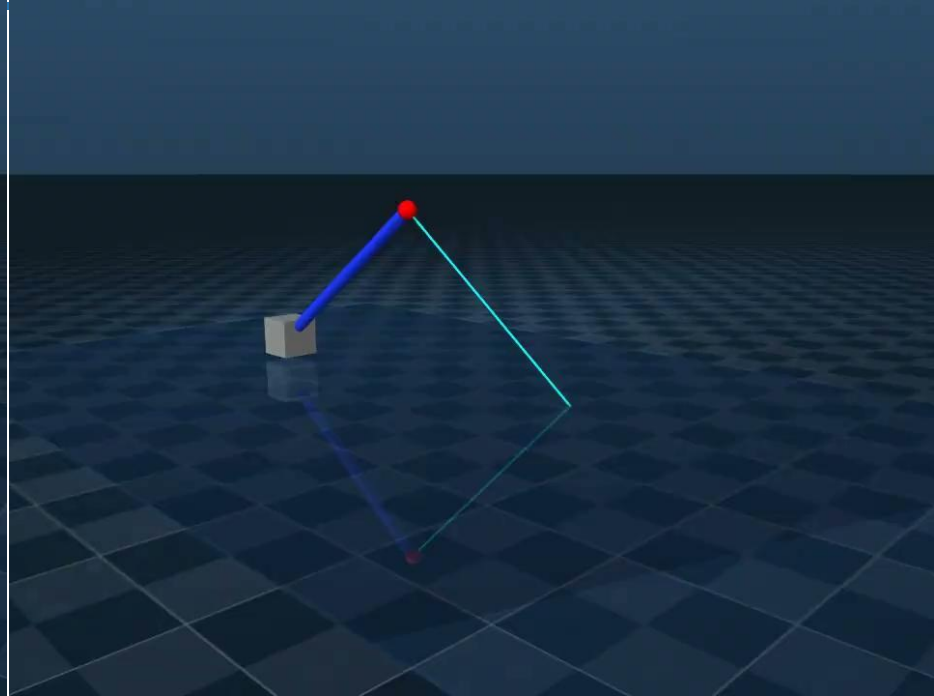


$$K_{f, \text{low/high}}(f_d - f_{\text{measured}}) + \theta_{\text{neutral}} = \theta_{\text{cmd}}$$

# Admittance Controller

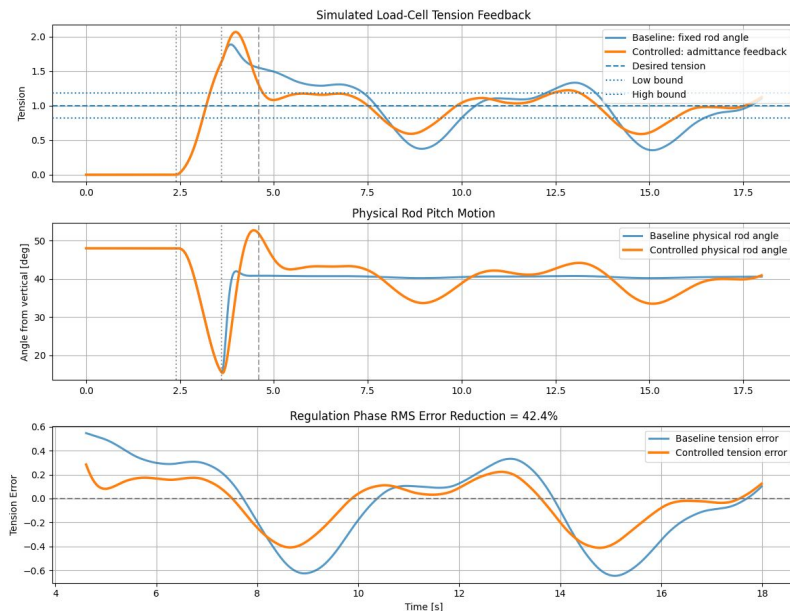


# Force Controller

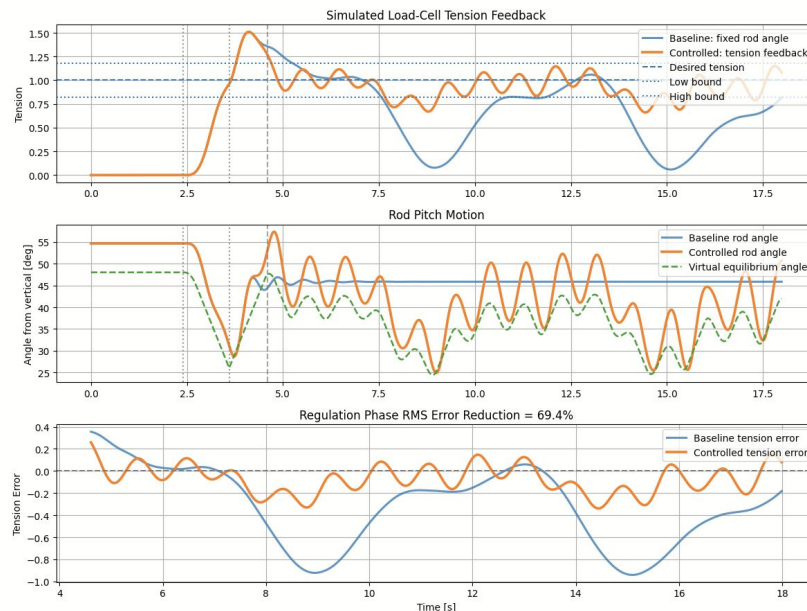


# Controller Comparison

## Admittance Control



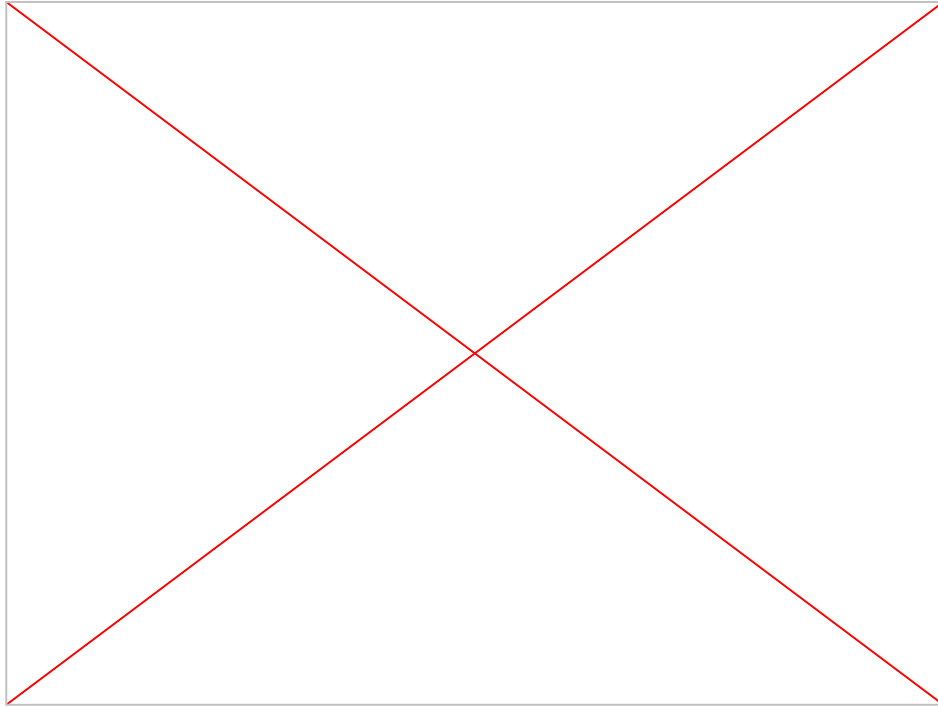
## Proportional Force Control



# Results

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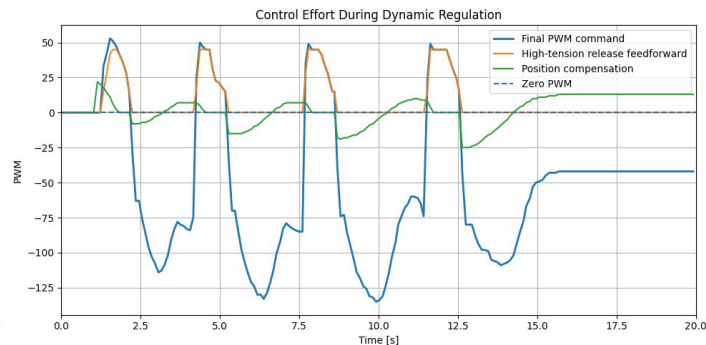
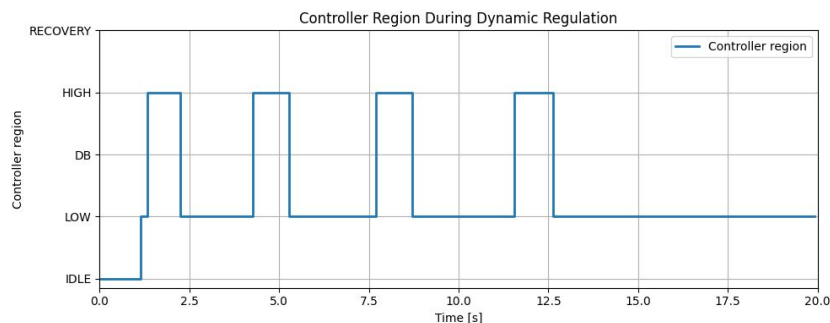


Manual pulling creates line-tension changes, and the rod responds through the Dynamixel motor.

# Dynamic Controller Response

**Test condition:** Rapid manual pulls were applied to create sudden line-tension changes.

**Key result:** the controller switched between release and pull-back actions, and the PWM command changed accordingly.



# Post-Transient Regulation Accuracy

## Test condition:

Initial manual pull followed by steady holding to evaluate whether the controller can return the tension near the 3.5 N target.

Target tension: 3.5 N

Final 5 s window:

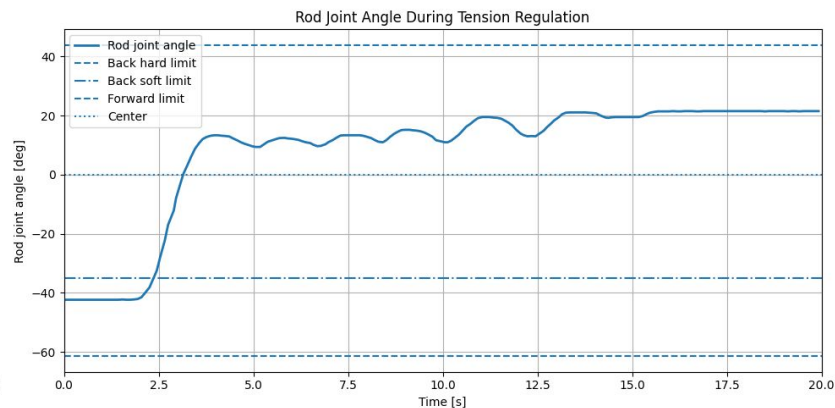
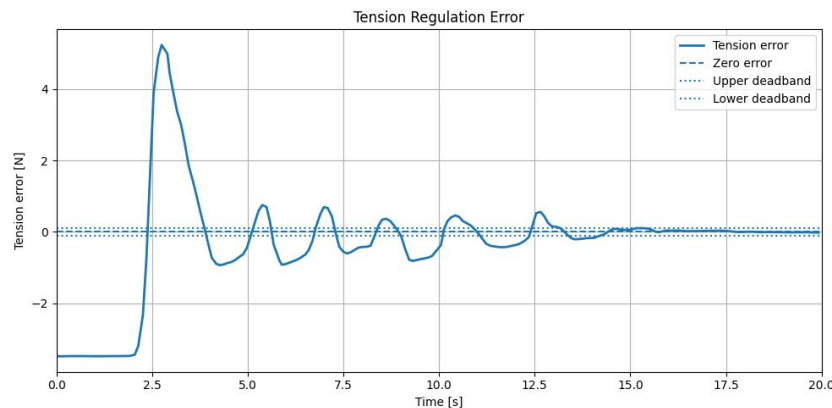
Mean tension: 3.52 N

MAE: 0.025 N (0.72%)

RMSE: 0.036 N (1.03%)

## Takeaway:

After the initial engagement and correction phase, the controller reduced the tension error and maintained the measured tension close to the target



# Future Developments

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# Hardware Developments

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## **Redundantly actuated serial joint**

- Emulate rod compliance with active control capability?
- Macro-micro manipulator structure
  - the heavy base joint handle low-frequency, high-amplitude positioning (like tracking boat's movement)
  - A smaller, redundantly actuated serial joint near the tip can be tuned with a much higher control loop frequency to handle high-frequency stochastic disturbances
- Null space optimization
  - infinite joint configs.
  - initiate secondary optimization objectives (torque minimization)
- Dynamic impedance shaping
  - redundant motor acts as dynamically tunable physical stiffness unit that adheres to control needs

## **A compliant segment** added to the end of the rod

- tune stiffness characteristics
- mechanical LPF, dealing with incredibly jerky fish motion in true fishing scenarios
- bumpless transfer stabilization (state transitions), reduce risky torque spikes

# Hardware testing rig

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- Actuatable object to mimic a fish that has planarized movement capabilities to perform varied experiments on our hardware system and its accompanying controller design
- Adding more DOF
- Flexible rod

# Practical Additions

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Bringing the product to 'market'

- Tensioning reel mechanism
- actuated hooking system

# Algorithmic extensions

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## More robust state switching capability

Finite State machine

Tension break detection and funnel synthesis

Admittance Control hardware implementation

## 2D Tension control

Hybrid control?

## Adaptive neural network

Augmentation of control loop with an adaptive neural network feedforward term that learns the stochastic frequency of the fish's pulling behavior, effectively predicting the required  $\dot{q}_d$  before the tension error grows too large.

**Sim-to-real:** Using the simulation as an AI training tool to optimize fish tensioning behavior.

- Direct RL: neural network
- Layered RL: controller parameter tweaking



# Thank You

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# Sources

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<https://www.researchgate.net/publication/344142164> Haptic Display Using Fishing Rod